34 An electric current of 12 A is in a wire of length 0.35 m.

The average drift speed of the free electrons (charge carriers) in the wire is $5.0 \times 10^{-4}\,\text{m s}^{-1}$.

How many free electrons are in the wire?

A 1.3×10^{16} **B** 5.3×10^{22} **C** 1.5×10^{23} **D** 4.3×10^{23}